involving liquid propulsion. Citing several references, I impressed upon Sarabhai that several missiles were powered by liquid fuel engines. I told him that while India can build its Intemediate-Range Ballistic Missiles (IRBMs) with solid, Inter-Continental Ballistic Missiles (ICBMs) should be powered by liquid fuel. Similarly, I told him, if we are to graduate from sounding rockets to satellite launchers, we have to develop liquid propulsion systems. Sarabhai was convinced.

Kalam, meanwhile, was working on a satellite launch vehicle with solid fuel as the propellant. It later came to be known as SLV-3. While I respected Kalam's dedication to his pet subject—the man literally lived at the workplace, always studying models and putting together solid boosters—I couldn't agree with his eagerness to dismiss liquid propellants as the future. It was a clear solid versus liquid tussle. The US, meanwhile, was planning to land man on moon.

Kalam found a useful friend in M R Kurup who was heading the Rocket Propellant Plant (RPP), which made only solid fuel. Ironically, I, the sole proponent of liquid propulsion, was stuck in the Rocket Engineering Division headed by Kalam.

Parallel to this, another group was pitted against Kalam. It comprised Y Janardhana Rao, V R Gowarikar, Subhash Chandra Gupta, D S Rane, Muthunayagam, Amba Rao, M K Mukherjee and a few others. All of them where PhDs and handpicked by Sarabhai, making it a potent group. But the problem was the eminent doctors did not know how to put a rocket together. The group had scant respect for Kalam and Murthy, who had only a bachelor's degree in engineering. That made me also think how I would survive, leave alone flourish as a scientist, only with a graduation in engineering.

I had wondered why Sarabhai, such a visionary, randomly brought people without assigning a clear job profile. It would become clearer to me soon when Sarabhai himself would open up on his experiment: Grab the best of brains, mostly people who have earned PhDs abroad, and keep them in India. Let them do whatever they want in space science. A structure will emerge sooner or later.